

Differentiate White, Black and Grey Hat Hackers

Factor	White Hat	Black Hat	Grey Hat
Legal permission	Has permission	No permission	Mixed permission
Intent	Improve security	Personal gain	Notify owner
Methods	Legal methods	Illegal methods	Both legal/illegal
Disclosure	Reports findings	Hides findings	May disclose
Motivation	Ethical defense	Malicious attack	Curiosity/fun
Penalty risk	No penalty	Heavy penalty	Occasional penalty
Example	Certified tester	Data thief	Hacks then informs
Law alignment	Fully legal	Fully illegal	Between laws

Compare active attacks vs Passive attacks

Factor	Active Attacks	Passive Attacks
Goal	Modify system	Monitor/steal data
Detection	Easy to detect	Hard to detect
System impact	Disrupts operations	No disruption
Example	DoS, spoofing	Sniffing, eavesdrop
Data alteration	Yes	No
Resource use	High	Low
Countermeasure	Firewalls, IDS	Encryption
Legality	Clearly illegal	Gray area

Explain Hackers, Crackers, Phreakers

Factor	Hackers	Crackers	Phreakers
Primary focus	Systems/networks	Breaking software	Phone networks
Intent	Explore/fix	Steal/damage	Free calls
Ethics	Often ethical	Always unethical	Mixed ethics

Factor	Hackers	Crackers	Phreakers
Skill level	High programming	High programming	Telecom specialist
Legal status	Mostly legal	Fully illegal	Often illegal
Origin term	Positive (MIT)	Negative spin-off	1970s phone culture
Modern example	Security researcher	Software pirate	VoIP manipulator
Key tool	Scripts/code	Cracking tools	Blue/red boxes

Difference between hacking and ethical hacking with examples

Factor	Hacking	Ethical Hacking
Permission	Unauthorized	Authorized
Legality	Illegal	Legal
Intent	Malicious/curious	Defensive/improve
Reporting	No report	Full report prepared
Example 1	Stealing credit cards	Bank security test
Example 2	Defacing website	Hospital penetration test
Example 3	Installing ransomware	Government system audit
Outcome	Harm/loss	Security fix

Differentiate between Vulnerability Assessment and Penetration Testing

Factor	Vulnerability Assessment	Penetration Testing
Objective	List weaknesses	Exploit weaknesses
Depth	Surface scan	Deep exploitation
Automation	Mostly automated	Mostly manual
Report type	Vulnerability list	Proof-of-concept attacks
False positives	Common	Rare
Time required	Hours to days	Days to weeks
Risk to system	Very low	Moderate risk

Factor	Vulnerability Assessment	Penetration Testing
Final result	Prioritized vulnerabilities	Actual breach paths